

Exercise 2: Statistics + Parameter Estimation
(to be returned before November 4th, 8:15)

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In this exercise, we repeat some tools from statistics and solve a simple estimation problem using linear least squares.

Exercise Tasks

1. ON PAPER: The covariance matrix of a vector-valued random variable $X \in \mathbb{R}^n$ with mean $\mathbb{E}\{X\} = \mu_X$ is defined by

$$\text{cov}(X) := \mathbb{E}\{(X - \mu_X)(X - \mu_X)^\top\}.$$

Prove that the covariance matrix of a vector-valued variable $Y = AX + b$ with constant $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ is given by

$$\text{cov}(Y) = A \text{cov}(X) A^\top.$$

(2 points)

2. ON PAPER: Let $X \in \mathbb{R}^n$ be a vector-valued random variable with mean $\mu \in \mathbb{R}^n$. Show that the covariance matrix $\text{cov}(X)$ can also be calculated by

$$\text{cov}(X) = \mathbb{E}\{XX^\top\} - \mu\mu^\top$$

(2 points)

3. ON PAPER: Suppose we are measuring a constant $x_0 \in \mathbb{R}$ perturbed by random independent noise ϵ with mean $\mu_\epsilon = 0$ and variance $\sigma_\epsilon^2 > 0$, i.e. we have

$$x = x_0 + \epsilon.$$

- (a) State the mean μ_x and the variance σ_x^2 of the random variable x . (1 point)
- (b) Let $x(n) = (x_1, \dots, x_n)$ denote a sample of n observations of x . The sample mean is given by $\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$ and it is an unbiased estimator of the mean μ_x . What is the variance of $\bar{x}(n)$? (1 point)
- (c) Prove that the Least Squares (LS) estimate for x_0 is the sample mean $\bar{x}(n)$. State the minimization problem explicitly. Is it convex? (2 bonus points)

4. Consider the following experimental setup, where we measure the temperature-dependent expansion of a fluid in a long transparent pipe, such as in a traditional thermometer. We describe the length of the visible fluid with the affine model

$$m(T; \theta_1, \theta_2) = \theta_1 \cdot T + \theta_2. \quad (1)$$

where T is the temperature in Celsius, and the parameters θ_1 and θ_2 relate to the specific expansion coefficient of the material and the length of the fluid at temperature $T = 0^\circ\text{C}$, respectively. Below, you find the measurements. Using the data, you will compute estimates for the parameters θ_1 and θ_2 .

k	1	2	3	4
$T(k)$ [$^\circ\text{C}$]	5	15	35	60
$L(k)$ [cm]	6.55	9.63	17.24	29.64

- (a) CODE: Plot the measurements $T(k), L(k)$ using 'x' markers. (0.5 points)
- (b) ON PAPER: Using the model from above, calculate the experimental values for the parameters θ_1 and θ_2 by minimizing the sum of squared distances, i.e.

$$\theta_1^*, \theta_2^* = \arg \min_{\theta_1, \theta_2} \sum_{k=1}^4 (L(k) - m(T(k); \theta_1, \theta_2))^2,$$

Give an analytical expression for the values of θ_1^* and θ_2^* with respect to the measurements $T(1), \dots, T(4)$ and $L(1), \dots, L(4)$.

Hint: Compute the solution by setting the gradient of the objective function $f(\theta_1, \theta_2) = \sum_k (L(k) - m(T(k), \theta_1, \theta_2))^2$ with respect to the parameters (θ_1, θ_2) to zero, i.e. $\nabla f(\theta_1, \theta_2) = 0$. This will give you a 2×2 linear system. Check if the objective function is convex!

CODE: Calculate the values of θ_1^* and θ_2^* using the data. Plot the fit $m(T; \theta_1^*, \theta_2^*) = \theta_1^* T + \theta_2^*$ over the range $[0, 100]$ in the same figure as before. (2 points)

- (c) CODE: Now, use a third order polynomial and fit it to the data using `np.polyfit`. Again minimize the sum of squared distances to find optimal values for the coefficients of your model equation. Plot the fit in the same figure as before. (0.5 point)
- (d) CODE: You take another measurement: At $T = 70^\circ\text{C}$ you measure a length of $L = 32.89$ cm. You can use this additional datapoint to validate your fit. Add the measurement to the existing plot.

ON PAPER: Which fit looks more reasonable to you?

Hint: The phenomenon of fitting a model to a data set which then does not pass validation is called 'overfitting'. (1 point)

This sheet gives in total 10 points and 2 bonus points.

1) given $Y = AX + b$
 with $A \in \mathbb{R}^{m \times n}$ & $b \in \mathbb{R}^m$
 proof: $\text{cov}(Y) = A \text{cov}(X) A^T$

$$\text{cov}(X) = E\{(X - \mu_X)(X - \mu_X)^T\}$$

$$\text{cov}(Y) = E\{(Y - \mu_Y)(Y - \mu_Y)^T\}$$

$$= E\{(Y - E\{Y\})(Y - E\{Y\})^T\}$$

$$= E\{(AX + b - A\mu_X - b)(AX + b - A\mu_X - b)^T\}$$

$$= E\{(AX - A\mu_X)(AX - A\mu_X)^T\}$$

$$= E\{A(X - \mu_X)[A(X - \mu_X)]^T\}$$

$$= E\{A \underbrace{(X - \mu_X)(X - \mu_X)^T}_{\text{cov}(X)} A^T\}$$

$$= E\{A\} E\{(X - \mu_X)(X - \mu_X)^T\} E\{A^T\}$$

$$= A \text{cov}(X) A^T$$

$$\text{where } E\{Y\} = E\{AX + b\}$$

$$= A E\{X\} + b$$

$$= A \mu_X + b$$

$$Y = AX + b$$

• Expectation E is linear

& $E\{\text{constant}\} = \text{constant}$.

Q2: given $X \in \mathbb{R}^n$, $\mu \in \mathbb{R}^n$

prove: $\text{cov}(X) = E\{XX^T\} - \mu\mu^T$ ($\mu = \mu_X$: for short)

$$\text{cov}(X) = E\{(X - \mu)(X - \mu)^T\}$$

$$= E\{(X - \mu)(X^T - \mu^T)\}$$

$$= E\{XX^T - X\mu^T - \mu X^T + \mu\mu^T\}$$

$$= E\{XX^T\} - E\{X\mu^T\} - E\{\mu X^T\} + E\{\mu\mu^T\}$$

$$= E\{XX^T\} - \mu\mu^T - \mu\mu^T + \mu\mu^T$$

$$= E\{XX^T\} - \mu\mu^T$$

$$\begin{array}{l|l} X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} & X\mu^T = (n \times n) \text{ matrix} \\ \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} & \mu^T X = (n \times n) \text{ matrix} \end{array}$$

$$E\{X\} = \mu$$

$$E\{X^T\} = \mu^T$$

Q₃: given: $x = x_0 + \varepsilon$ $\left\{ \begin{array}{l} x_0 \in \mathbb{R} \rightarrow x_0: \text{truth} \\ \mu_\varepsilon = 0 \\ \sigma_\varepsilon^2 > 0 \end{array} \right\} \rightarrow \varepsilon: \text{noise.}$

a) μ_x is the mean of x followed by this expectation $E\{x\}$

• $E\{x\} = E\{x_0 + \varepsilon\} = E\{x_0\} + E\{\varepsilon\} = x_0 + 0$

$\Rightarrow \boxed{\mu_x = x_0}$ $\rightarrow$ this one uses for (b)

$\sigma_\varepsilon^2 = E\{(\varepsilon - \mu_\varepsilon)^2\}$
 $= \boxed{E\{\varepsilon^2\}}$

• $\sigma_x^2 = E\{(x - \mu_x)^2\} = E\{(x - x_0)^2\} = E\{(x_0 + \varepsilon - x_0)^2\}$

$= E\{\varepsilon^2\} = \boxed{\sigma_\varepsilon^2}$ (variance of the noise)

b) $\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$, with $x(n) = (x_1, \dots, x_n)$ $\rightarrow$ n samples.

$\text{Var}(\bar{x}) = \sigma_{\bar{x}}^2 = E\{(\bar{x} - \mu_{\bar{x}})^2\}$

• Expectation of a sum is a sum of expectations.

• $\mu_{\bar{x}} = E\{\bar{x}(n)\} = E\left\{\frac{1}{n} \sum_{i=1}^n x_i\right\}$

$= \frac{1}{n} \sum_{i=1}^n E\{x_i\}$

from a: $\boxed{E\{x_i\} = x_0}$

$= \frac{1}{n} \sum_{i=1}^n x_0 = \frac{n \cdot x_0}{n} = \boxed{x_0}$

$x_i = x_0 + \varepsilon$

• $\sigma_{\bar{x}}^2 = E\{(\bar{x} - \mu_{\bar{x}})^2\} = E\left\{\left[\left(\frac{1}{n} \sum_{i=1}^n x_i\right) - x_0\right]^2\right\}$

$= E\left\{\left[\left(\frac{1}{n} \sum_{i=1}^n x_0 + \varepsilon(i)\right) - x_0\right]^2\right\} \mid \varepsilon \text{ is not a constant}$

$= E\left\{\left[\underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_0\right)}_{x_0} + \frac{1}{n} \sum_{i=1}^n \varepsilon(i) - x_0\right]^2\right\}$

$= E\left\{\left(x_0 + \frac{1}{n} \sum_{i=1}^n \varepsilon(i) - x_0\right)^2\right\}$

$$= E \left\{ \left(\frac{1}{n} \sum_{i=1}^n \epsilon(i) \right)^2 \right\} = \frac{1}{n^2} \cdot E \left\{ \left(\sum_{i=1}^n \epsilon(i) \right)^2 \right\}$$

The term: $\left(\sum_{i=1}^n \epsilon(i) \right)^2 = [\epsilon(1) + \epsilon(2) + \dots + \epsilon(n)]^2$

$$= \underbrace{(\epsilon^2(1) + \epsilon^2(2) + \dots + \epsilon^2(n))}_{\text{square term}} + \underbrace{(2\epsilon(1)\epsilon(2) + 2\epsilon(1)\epsilon(3) + \dots)}_{\text{cross term}}$$

$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$
 $(a+b+\dots+n)^2 = a^2 + b^2 + 2ab + \dots$

→ Look at the cross term: $E \{ (2\epsilon(1)\epsilon(2) + 2\epsilon(1)\epsilon(3) + \dots) \}$

$$= 2E \{ \epsilon(1)\epsilon(2) + \epsilon(1)\epsilon(3) + \dots \}$$

$$= 2E \{ \epsilon(1) \} E \{ \epsilon(2) \} + E \{ \epsilon(1) \} E \{ \epsilon(3) \} + \dots$$

ϵ is noise $\rightarrow E \{ \epsilon(i) \} = 0$

→ $\sigma^2_{\bar{x}} = \frac{1}{n^2} E \{ \epsilon^2(1) + \epsilon^2(2) + \dots + \epsilon^2(n) \}$ (from a

$$E \{ \epsilon^2 \} = \sigma_e^2$$

→ $\sigma^2_{\bar{x}} = \frac{1}{n^2} (\sigma_{\epsilon(1)}^2 + \sigma_{\epsilon(2)}^2 + \dots + \sigma_{\epsilon(n)}^2)$

Because the noise is identically distributed $\rightarrow \sigma_{\epsilon(1)}^2 = \dots = \sigma_{\epsilon(n)}^2 = \sigma_e^2$

⇒ $\sigma^2_{\bar{x}} = \frac{1}{n^2} \cdot n \cdot \sigma_e^2 = \frac{\sigma_e^2}{n}$ (the variances for all measurements are the same).

→ This proves that when taking the measurements multiple times, the larger n — "Number of measurements", the smaller the variance — "uncertainty".

c). called: $f(x)$ is the objective function for this optimization problem

$$f(x) = \sum_{i=1}^n [x(i) - x_0]^2$$

⇒ optimized value is x^* subject to argmin $f(x)$

$$x^* = \arg \min \sum_{i=1}^n (x(i) - x_0)^2$$

• prove that $\arg \min \sum_{i=1}^n (x(i) - x_0)^2 = \bar{x}(n)$. [sample mean].
 $\hookrightarrow \boxed{\mu_{\bar{x}} = x_0}$ proved above

- Take the derivative of $\sum_{i=1}^n (x(i) - x_0)^2$.

$$\frac{d}{dx_0} f(x) = \frac{d}{dx_0} \sum_{i=1}^n [x(i) - x_0]^2 = \sum_{i=1}^n \frac{d}{dx_0} [x(i) - x_0]^2$$

$$= \sum_{i=1}^n -2[x(i) - x_0]$$

$$\bullet \text{ let } \frac{d}{dx_0} f(x) = 0 \Leftrightarrow \sum_{i=1}^n -2[x(i) - x_0] = 0$$

$$\Leftrightarrow \sum_{i=1}^n [x(i) - x_0] = 0$$

$$\Leftrightarrow \sum_{i=1}^n x(i) - \sum_{i=1}^n x_0 = 0$$

$$\Leftrightarrow \sum_{i=1}^n x(i) = n \cdot x_0 = 0$$

$$\Leftrightarrow \sum_{i=1}^n x(i) = n \cdot x_0$$

$$\Leftrightarrow \boxed{\frac{1}{n} \sum_{i=1}^n x(i) = x_0}$$

this is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x(i)$

$x(i) = x_0$ is wrong
 because this is the
 sum. $\sum x(i)$
 $= \sum x_0 \Rightarrow x(i)$ can
 vary around x_0 .

$$\boxed{\bar{x} = x_0}$$

• Take the second derivative

$$\frac{d}{dx_0} f'(x) = \frac{d}{dx_0} \sum_{i=1}^n -2(x(i) - x_0) = \sum_{i=1}^n \frac{d}{dx_0} [-2(x(i) - x_0)]$$

$$= \sum_{i=1}^n \left[\underbrace{\frac{d}{dx_0} [-2(x(i))]}_0 + 2 \underbrace{\frac{d}{dx_0} x_0}_1 \right] = \sum_{i=1}^n +2 = +2 \cdot n$$

$\Rightarrow f''(x) = +2n > 0 \rightarrow x_0$ is the global minima of $f(x)$.

$\Rightarrow f(x)$ is strictly convex

Q 4 : b) . $m(T, \theta_1, \theta_2) = \boxed{\theta_1 T + \theta_2}$ estimator for pipe's length based on parameters θ_1 & θ_2

$$\theta_1^*, \theta_2^* = \arg \min_{\theta_1, \theta_2} \sum_{k=1}^4 \left(\underbrace{L(k)}_{\substack{\text{measured} \\ \text{pipe's length}}} - \underbrace{m(T(k); \theta_1, \theta_2)}_{\substack{\text{measured} \\ \text{temperature}}} \right)^2$$

$f(\theta_1, \theta_2)$

• Take the gradient of $f(\theta_1, \theta_2)$:

$$\begin{aligned} \frac{\partial f(\theta_1, \theta_2)}{\partial \theta_1} &= \frac{\partial}{\partial \theta_1} \sum (L(k) - (\theta_1 T(k) + \theta_2))^2 \\ &= \sum \frac{\partial}{\partial \theta_1} \underbrace{[L(k) - (\theta_1 T(k) + \theta_2)]^2}_{u} = u' \cdot u \\ &= \sum \left[2 [L(k) - \theta_1 T(k) - \theta_2] [-T(k)] \right] \\ &= -2 \sum (L(k) - \theta_1 T(k) - \theta_2) \cdot T(k) \end{aligned}$$

$$\frac{\partial f(\theta_1, \theta_2)}{\partial \theta_1} = 0 \Rightarrow \cancel{\sum} [L(k) T(k) - \theta_1 T(k) \cdot T(k) - \theta_2 T(k)] = 0$$

$$\Leftrightarrow \sum L(k) T(k) - \theta_1 \sum T(k) \cdot T(k) - \theta_2 \sum T(k) = 0$$

With $\theta_2 = \bar{L} - \theta_1 \bar{T} \rightarrow$ proved below

$$\Leftrightarrow \sum L(k) T(k) - \theta_1 \sum T(k)^2 - \bar{L} \sum T(k) + \theta_1 \bar{T} \sum T(k) = 0$$

$$\Leftrightarrow \theta_1 [\bar{T} \sum T(k) - \sum T(k)^2] = \bar{L} \sum T(k) - \sum L(k) T(k)$$

$$\theta_1^* = \frac{\bar{L} \sum T(k) - \sum L(k) T(k)}{\bar{T} \sum T(k) - \sum T^2(k)}$$

$$\begin{aligned} \frac{\partial f(\theta_1, \theta_2)}{\partial \theta_2} &= \sum \frac{\partial}{\partial \theta_2} \left[L(k) - (\theta_1 T(k) + \theta_2) \right]^2 = u \cdot u \\ &= \sum \left[2 [L(k) - \theta_1 T(k) - \theta_2] \cdot (-1) \right] \\ &= -2 \sum (L(k) - \theta_1 T(k) - \theta_2) \end{aligned}$$

$$\text{Let } \frac{\partial f(\theta_1, \theta_2)}{\partial \theta_2} = 0 \Rightarrow -2 \sum (L(k) - \theta_1 T(k) - \theta_2) = 0$$

$$\Rightarrow \sum L(k) - \sum \theta_1 T(k) - \sum \theta_2 = 0$$

$$\Rightarrow \sum L(k) - \theta_1 \sum T(k) = \sum \theta_2$$

$$\Rightarrow \sum L(k) - \theta_1 \sum T(k) = 4\theta_2$$

$$\Rightarrow \theta_2^* = \frac{1}{4} \sum L(k) - \frac{1}{4} \theta_1 \sum T(k)$$

$$\theta_2^* = \bar{L} - \theta_1 \cdot \bar{T}$$

• prove the convexity using Hessian Matrix.

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 f}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 f}{\partial \theta_2^2} \end{bmatrix}$$

$$\cdot \frac{\partial^2 f}{\partial \theta_1^2} = -2 \sum [L(k) - \theta_1 T(k) - \theta_2] \cdot T(k)$$

$$\cdot \frac{\partial^2 f}{\partial \theta_2^2} = -2 \sum (L(k) - \theta_1 T(k) - \theta_2)$$

$$\begin{aligned} \bullet \frac{\partial^2 f}{\partial \theta_1^2} &= \frac{\partial}{\partial \theta_1} \left(-2 \sum \left[\underbrace{L(k)}_0 - \theta_1 \underbrace{T(k)}_0 - \underbrace{\theta_2}_0 \cdot T(k) \right] \right) \\ &= -2 \sum \frac{\partial}{\partial \theta_1} (-\theta_1 T(k)^2) \end{aligned}$$

$$= \boxed{2 \sum T(k)^2 \geq 0}$$

$$\bullet \frac{\partial^2 f}{\partial \theta_1 \partial \theta_2} = \frac{\partial}{\partial \theta_2} \left(-2 \sum \left[\underbrace{L(k)}_0 - \theta_1 \underbrace{T(k)}_0 - \underbrace{\theta_2}_0 \cdot T(k) \right] \right)$$

$$= \boxed{2 \sum T(k) \geq 0}$$

$$\bullet \frac{\partial^2 f}{\partial \theta_2 \partial \theta_1} = \frac{\partial}{\partial \theta_1} \left(-2 \sum \left(\underbrace{L(k)}_0 - \theta_1 T(k) - \underbrace{\theta_2}_0 \right) \right)$$

$$= \boxed{2 \sum T(k) > 0}$$

$$\bullet \frac{\partial^2 f}{\partial \theta_2^2} = \frac{\partial}{\partial \theta_2} \left(-2 \sum \left(\underbrace{L(k)}_0 - \theta_1 T(k) - \theta_2 \right) \right)$$

$$= \boxed{2 \sum 4 = 8 > 0}$$

$$\Rightarrow H = \begin{bmatrix} 2 \sum T(k)^2 & 2 \sum T(k) \\ 2 \sum T(k) & 8 \end{bmatrix} = 2 \begin{bmatrix} \sum T(k)^2 & \sum T(k) \\ \sum T(k) & 4 \end{bmatrix}$$

• H is positive-definiteness:

1. Top left element is positive: $\sum T(k)^2 > 0$

2. $\det(H) > 0 \Leftrightarrow 4 \sum T(k)^2 - \sum T(k)^2 = 3 \sum T(k)^2 > 0$

$$\Rightarrow \boxed{H \succ 0}$$

$$\Rightarrow \boxed{f(\theta_1, \theta_2) \text{ is convex}}$$

d) The one linear fit looks better because it is very close to the validation.

- The 3rd order polynomial fit is very good with measured data set, but the residual error between the fit and the validation is higher than linear fit $\rightarrow$ "overfitting"